INCH POUND

S-F-695G
INTERIM AMENDMENT 4 (SH)
7 April 1992
SUPERSEDING
INTERIM AMENDMENT 3
30 December 1987

FEDERAL SPECIFICATION

FRYERS, DEEP-FAT (ELECTRIC)

This interim amendment forms a part of Federal Specification S-F-695G, dated 31 January 1981, and is approved for use within the Naval Sea Systems Command, Department of the Navy.

PAGE 3

2.1, under "Federal Specifications": Delete "QQ-S-698 - Steel, Sheet and Strip, Low-Carbon".

2.1, under "Standards/Military": Add the following:

"MIL-STD-167/1 - Mechanical Vibration of Shipboard Equipment Type I - Environmental and Type II - Internally Excited.

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of

MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities."

FSC 7310

AMSC N/A
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PAGE 4

2.2, under "American Society for Testing and Materials (ASTM)": Add "A 366 - Steel, Carbon, Cold Rolled Sheet, Commercial Quality".

PAGE 5

3.4.1, line 2: Delete "QQ-S-698" and substitute "ASTM A 366".

PAGE 6

3.5, line 7, after "(submarine use): Add "The height of the type VI fryer from the deck to the top of the removable cover shall be + 1/2 inches."

PAGE 7

3.5.2.2.1, line 4, following "fryer." delete the period and:
Add "except that test switches which are used periodically to test the safety thermostat circuit may be located elsewhere provided they are located for safe and convenient access from the front of the fryer."

3.6.1.1, line 3, following "stainless steel": Add "or nickel plated steel".

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PAGE 10

3.6.8: Delete and substitute:

"3.6.8 Thermostat. An adjustable thermostat shall be provided with a minimum range from 200° to 275° F and a maximum range from 375° to 400° F. The thermostat shall have a positive "OFF" position and be marked at 50 degree intervals. A separate "ON-OFF" switch may be provided in addition to, or in lieu of, the positive "OFF" position to control power to the thermostat. Thermostat and switch shall be located so as to be readily accessible to the operator. The thermostat shall maintain an average temperature of 375 + 5° F when tested as specified in 4.4.2.1. The temperature amplitude of the thermostat shall not exceed 25° F when tested as specified in 4.4.2.2. Instructions for calibrating the thermostat shall be provided with each fryer (see 3.8.2). The thermostat shall limit the maximum temperature to 425° F."

3.6.8.1, line 6, following "4.4.3": Delete the sentence beginning "In lieu of" and insert: "A means shall be provided to facilitate checking the operation of the over-temperature thermostat by rendering the first thermostat inoperative. A signal light shall indicate when the overtemperature thermostat is operated when tested as specified in 4.4.3."

PAGE 11

3.6.8.1 (cont'd), line 8: Delete "not exceeding 430° F" and substitute "of 430° F through 460° F".

3.6.8.1.1: Delete first sentence and substitute: "For Naval Shipboard use (use 6.2), the deep fat fryer shall be provided with a primary thermostat shunt bypass assembly."

PAGE 12

3.6.9.3, first sentence, following "above": Delete "and shall be capable of recovery to 375° F prior to completion of frying cycle".

PAGE 14

3.9, line 2: Delete "dimensional requirements for fryers shall be as specified in figures 1, 2, and 3 and".

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3.9.2: Delete and substitute:

"3.9.2 Legs and mountings. The bottom of the legs for the type I, size 1, type II and type VI fryers shall be provided with holes for the mountings. The type I fryer shall have 3/8 - 16 threaded leg mounting holes. The type II and type VI fryers shall have 9/16 inch diameter leg mounting holes for 1/2 inch diameter bolts. All legs shall be fabricated of stainless steel and the thickness shall not be less than 16 gauge. Legs shall be reinforced without cross bracing to form a rigid assembly."

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* Add new paragraph:

"3.9.3.1 High voltage labels. When specified for shipboard use (see 6.2), a Danger High Voltage label shall be affixed to the equipments outer case assembly, on or adjacent to each service access cover and adjacent to one of the fasteners which secures the cover. A high voltage warning label shall also be placed near the high voltage components inside the equipment. The label shall include but is not limited to the following warnings:

a. A warning of high voltage.
b. Power supply must be disconnected before using.
c. Access covers must be in place during use.
d. Servicing should be done by authorized individuals."

PAGE 15

3.9.8, after last sentence add:

"The deep fat fryers shall be insulated such that the exterior surface temperature does not exceed 140° F. Heat-resisting hand grips shall be provided in order to insulate any handlers such that the surface temperature of the grip does not exceed 120° F."

Add the following new paragraphs:

"3.9.11 U.L. requirement. The unique requirement for fryers to be connected to a remote shunt trip disconnect (see 3.6.8.1) is recognized to preclude overall U.L. approval of the complete fryer. In lieu thereof, the fryer shall be constructed of U.L. approved components and in such a manner that the fryer could be U.L. approved upon incorporation of an integral over temperature thermostat control system. When style 2 swing-up heating elements are supplied in fryers, the thermostatic controls may be mounted in the tilting head."

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"3.9.12 Electromagnetic compatibility. When specified (see 6.2), the units shall be designed and equipped for electromagnetic capability in accordance with the requirements of MIL-STD-461, class A4 for surface ships and class A5 for submarines. The equipment shall meet the emission and susceptibility requirements for CE01, CE03, and RE02 (see 4.4.11)."

PAGE 16

Add the following paragraph:

"3.15 Human Factors Criteria. Human factors engineering criteria, principles and practices, as defined in MIL-STD-1472, shall be incorporated into the design of the deep fat fryer. The equipment shall permit safe and efficient maintenance and operational tasks by the 5th percentile female to the 95th percentile male as defined in sections 5.6 and 5.9 of MIL-STD-1472. Sufficient clearance or free area shall be provided around an item so that an individual with applicable body dimensions and physical capabilities can safely operate, maintain, remove or replace that item. When establishing accessibility requirements, both physical and visual access must be provided, along with access for the use of any tools, test equipment, or repair parts. Controls or switches needed for operation or maintenance shall be selected and integrated into the machines so as to meet the applicable requirements of sections 5.2, 5.4 and 5.9 of MIL-STD-1472. When inspecting for defects and performing tests (see section 4), the equipment shall adhere to the human factors engineering considerations listed herein."

PAGE 17

Table IV, under "Examine" and following "Signal light or lights": Delete listed defect and substitute "Not provided (power "ON" and "overtemperature thermostat" lights)".

PAGE 18

4.2.2.3, line 2, after "and": Add "when required for shipboard use."

PAGE 19

4.4.3: Delete the second and third sentences and substitute: "The first thermostat shall be rendered inoperative in accordance with the manufacturer's operating instructions. The hydrogenated vegetable oil temperature shall be increased until the overtemperature thermostat specified in 3.6.8.1 opens the heating element circuit."

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4.4.3 (cont'd), line 4, after "times": Add a period and delete the remainder of the sentence.

Line 5: Delete "sall" and substitute "shall".

4.4.4.2(a) last sentence: Delete.

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4.4.9: Delete entire paragraph and substitute the following:

"(a) Shipboard vibration test. The griddle under normal operating conditions, shall be tested in accordance with MIL-STD-167/1, Type I equipment. The griddle shall be secured to the test machine in the same manner that it will be secured on shipboard (see 3.9.2). Failure of the griddle to perform its function during and after testing, or meeting requirements of 3.9.5 shall constitute failure of this test."

Add the following new paragraph:

"4.4.11 Electromagnetic interference test. When required, the units shall be tested by the contractor in accordance with test methods CE01, CE03, and RE02 of MIL-STD-462. The contractor shall furnish written certification that the equipment meets MIL-STD-461 requirements. Nonconformance with the requirements specified shall constitute failure of the test.

PAGE 23

6.2, (i) line 1, after "3.6.8.1": Add ", 3.6.8.1.1".

Add the following new paragraph:

"6.2.2 Technical manuals. The requirement for technical manuals should be considered when this document is cited on a contract. If technical manuals are required, a contract exhibit must be prepared to fully describe statement of work criteria and delivery instructions, and cite the applicable technical manual. The technical manuals must be acquired by separate Contract Line Item Number (CLIN) in the contract."

PAGES 24 and 25

Delete Figures 1, 2 and 3.

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